AI Incident Report Assistant: On-Device, Voice-Driven Incident Reporting with Department-Specific Forms

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Abstract

We propose a voice assistant that runs entirely on the Qualcomm dev kit and turns speech into a department-specific incident report. Streaming ASR turns English, Hindi or Hinglish speech into English text. A liquid neural network (LNN) on the Hexagon NPU drops off-topic sentences, so the concurrently running on-device LLM extracts facts only from relevant speech and calls a fixed set of local functions. Laya, a non-autoregressive decision model quantized to INT8, then picks the incident type in one forward pass with probabilities. It generates no text, so it cannot invent a category. Local configuration maps each of 28 incident types to one of 6 departments, its required fields and its priority rules. The app asks only for what is missing, safety first; the user reviews and corrects, and the report lands in the right department's queue and on the floor-plan map. No cloud API is used at any point.

1 Problem Statement and Our Understanding

The problem as given. Build end-to-end, voice-driven incident reporting with on-device generative AI and function calling, following SPEAK → UNDERSTAND → EXTRACT → REVIEW → SUBMIT. An employee talks instead of typing into a portal. The app fills the report fields (type, severity, location, time, persons, equipment, description, actions, witnesses, follow-up), asks for missing mandatory fields, then shows the form for review and submission.

Evaluation criteria. All models on-device with Snapdragon acceleration and no cloud APIs; natural conversational speech; ASR accuracy; extraction accuracy; correct mapping to report fields.

Our understanding of what is actually hard.

- *Different incidents need different forms* and different teams. A leak, a lost badge and a chemical spill share almost no fields.
- *Never invent a value.* A guessed location or a missed hazard is worse than a blank, and "I don't know" must be a valid answer.
- *People ramble.* Small talk must not reach extraction.
- *It must feel like a conversation:* few short questions, safety first, within an on-device latency budget, in English, Hindi or Hinglish.

2 Distinguishing Contributions

The system works out *which* incident it is hearing before deciding what to ask next, and never leaves the chip to do it. Specifically:

(1) Two-tier neural pipeline. A tiny LNN (≈0.9 ms on the Hexagon NPU) screens every sentence as it streams, so the LLM running alongside it only processes relevant speech; Laya then classifies in one pass.

(2) Real function calling. The LLM itself decides when to resolve a room name, check a zone's incident history or load a form, from a fixed allowlist, against the local database, with every call logged. Priority comes from fixed P1–P4 rules, not a model's guess.

(3) Department-aware, incident-specific forms. 28 incident types, each with its own form and department. One answer fills every field it touches, and nothing already answered is asked again.

(4) Bilingual by design. English, Hindi and code-mixed speech, converted on-device into one clean English record any admin can read.

(5) Explicit uncertainty. Every field stores its evidence, its source and an explicit state: known, unknown or ambiguous. A correction invalidates everything derived from the old value, so the report stays consistent.

(6) A spatial view, not a ticket list. Each report highlights its zone on the floor plan with a pin where it was reported (not live employee tracking); one tap opens the full dossier.

(7) Zero cloud, verifiable live. Speech, understanding, extraction, the map and the dossier all run on the Snapdragon kit, demonstrated with the network disabled.

(8) Future work: relationship and trend analysis *(stretch, not in the core build).* A GNN flags reports that may share one cause, and a DynoTears-style background job mines incident history for recurring precursors. Both are bounded background jobs and are never shown as proven cause.

3 System Architecture and Pipeline

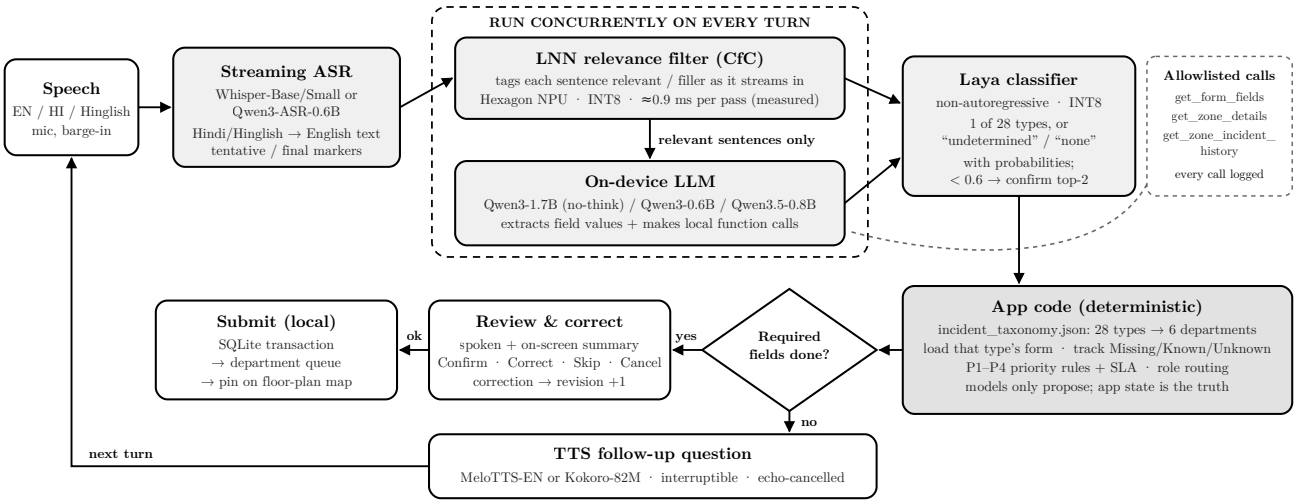


Figure 1. One reporting turn. The LNN and LLM run concurrently; the LLM only receives sentences the LNN marked relevant. All on the dev kit (Snapdragon 8 Elite SM8750P, QAIRT/SNPE 2.47); where several models are listed, the pick is made by on-kit benchmark.

3.1 Conversation Flow

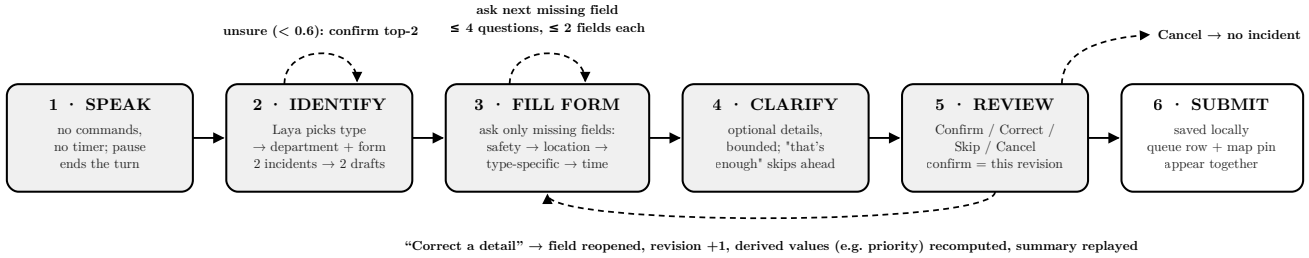


Figure 2. Conversation stages, mapped to the required SPEAK → UNDERSTAND → EXTRACT → REVIEW → SUBMIT flow. "I don't know" is stored as an explicit *Unknown* and never asked again. An unknown safety answer is treated as the worse case. The user can interrupt any spoken question.

3.2 Worked example: the problem statement's own input

"There was a water leak near Conference Room 2 around 10 AM. Maintenance has already isolated the area."

Laya: `water_leak` → Facilities Engineering · Plumbing Technician · default P3.
LLM (grounded): room = Conference Room 2 · time = around 10 AM · action taken = area isolated by maintenance.
Still missing: near_electrical (safety), water_spread (safety), leak_source, building, floor.
First question (safety first): "Is the water near any sockets or cables?"
Rules: if yes → **P1**, 15-min response, electrician and EHS also notified. If spreading → P2.

3.3 Model stack (all on-device)

Stage	Model	Status
ASR	Whisper-Base/Small (multilingual) vs Qwen3-ASR-0.6B	shortlist
Relevance filter	CfC liquid neural network, INT8, Hexagon NPU	benchmarked
Extraction + calls	Qwen3-1.7B (no-think) vs Qwen3-0.6B vs Qwen3.5-0.8B	shortlist
Incident type	Laya, quantized to INT8	selected
Speech output	MeloTTS-EN vs Kokoro-82M	shortlist
Storage	local SQLite; no network at run time	selected

3.4 Routing configuration: 28 incident types across 6 departments

Facilities Engineering: water leak, power outage, electrical hazard, HVAC, elevator
EHS: injury, fire/smoke, chemical spill, near miss
Housekeeping & Workplace: spill, washroom hygiene, pest, cafeteria
IT & Workplace Tech: network, conference room AV, printer
Global Security: badge reader, lost badge, tailgating, theft, CCTV
R&D Lab Operations: lab equipment, ESD, lab utilities

4 Planned Testing Methodology

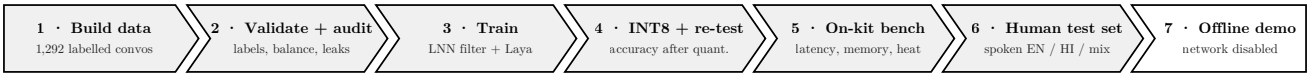


Figure 3. Test pipeline. A model build moves forward only when the previous gate passes.

4.1 Data and model accuracy

- **Dataset:** 1,164 training and 128 test conversations (≈3,200 user turns, ≈7,500 labelled sentences). It covers all 28 types plus *undetermined* and *none*, 168 look-alike conversations (e.g. spill vs chemical spill), 60 non-incidents and 40 two-incident reports. Labels: incident type per turn, relevant/filler per sentence, and fields mentioned.
- **No shortcuts:** automated audits check that the type can't be guessed from filler words, turn count, sentence length, location or writing style. The split is by conversation, and no test sentence appears in training.
- **Metrics:** incident-type accuracy (overall, on look-alike pairs and on "none"), relevance-filter precision and recall, and field-mention F1. All are re-measured after INT8 quantization.

4.2 Speech and language

- Held-out **spoken** recordings from team members in English, Hindi and Hinglish, with multiple speakers and real office noise. We check that room names, floors, times and equipment IDs survive Hindi → English conversion.
- Because the synthetic test set shares its authors with the training data, the human-recorded set is the number we trust.
- Open risk we will test: whether Hindi speakers understand the English follow-up questions.

4.3 End-to-end conversation quality

- **Field accuracy** and **false-fill rate** (a field marked known but wrong).
- **Repeated-question rate**: re-asking something already answered is a hard failure.
- **Hazard-question recall**: safety fields asked first whenever relevant.
- **Routing accuracy** (type → department), **completion rate** and turns to completion.
- Scripted cases: "I don't know", corrections at review, interruptions, two incidents in one report.

4.4 On-device performance (Snapdragon 8 Elite dev kit)

P95 target	Budget
Listening feedback on screen	< 100 ms
Stop speaking when user interrupts	< 150 ms
Next question, cached template	< 1 s
Next question, LLM-composed (e.g. disambiguation)	< 2 s

- Plus memory, power and thermal behaviour over several back-to-back reports. Already measured: LNN at 0.7–0.9 ms per pass on the Hexagon NPU (INT8, 5.1–5.5 MB).
- **Offline proof:** the network is switched off before the live demo, the code is checked for any non-local host, and all weights are preloaded.